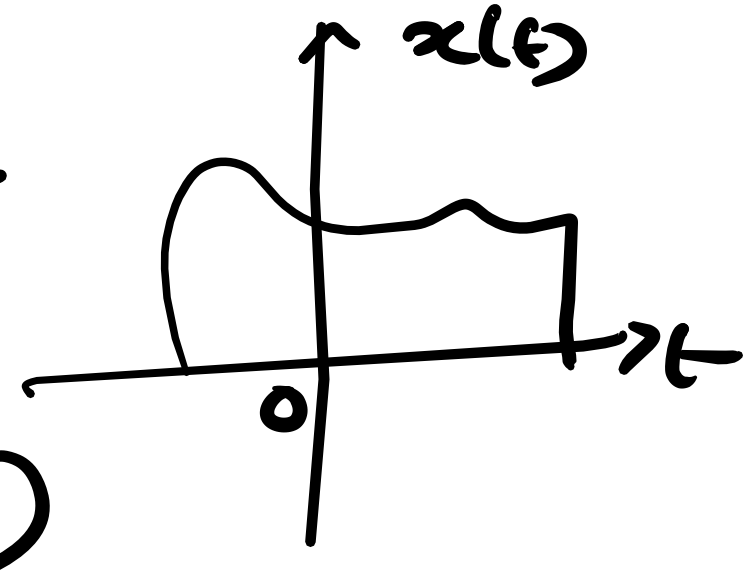
14/11/2025

L.T.

① For a right sided signal, ROC is right to the rightmost pole.

② For a left sided signal, ROC is left to the leftmost pole.

③ For a 2-sided signal, ROC will be the entire s-plane.
(and a finite duration signal)

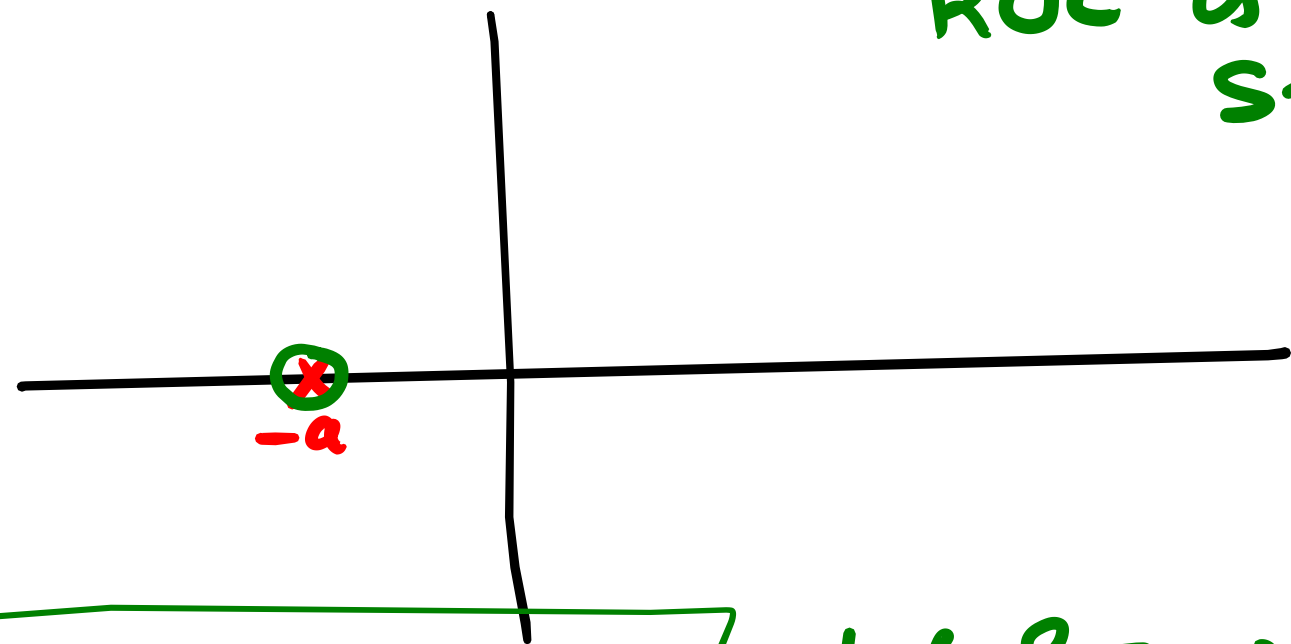


Ex $x(t) = \begin{cases} e^{-at}u(t) & 0 \le t \le T \\ 0 & \text{otherwise} \end{cases}$

$$X(s) = \frac{1}{s+a} - \frac{e^{-(s+a)T}}{s+a}$$

plot poles and zeros of $X(s)$.

ROC is the entire s-plane.



$$\lim_{s \to -a} X(s) = T$$

pole & zero at $s = -a$ cancel each other.

Ex

$$x(t) = t\,e^{-at}u(t)$$

$$X(s) = ?$$

$$X(s) = \int_{-\infty}^{\infty} x(t)\,e^{-st}\,dt$$

$$\frac{d}{ds}X(s) = \int_{-\infty}^{\infty} x(t)\,(-t)\,e^{-st}\,dt$$

$$= \int_{-\infty}^{\infty} (-t\,x(t))\,e^{-st}\,dt$$

$$x(t) \rightleftharpoons X(s) \qquad ROC = R$$

$$-t\,x(t) \rightleftharpoons \frac{d}{ds}X(s) \qquad ROC = R$$

$y(t) = e^{-at}u(t)$

$Y(s) = \frac{1}{s+a}$

R: Re{s} > -a

$\frac{d}{ds}Y(s) = \frac{-1}{(s+a)^2}$

$\Downarrow$

$+t\,e^{-at}u(t)$

$\Uparrow$

$+\frac{1}{(s+a)^2}$

ROC = R

$$x(t) = t^2 e^{-at} u(t)$$

$$X(s) = \frac{2}{(s+a)^3} \qquad \text{ROC} : R$$

$$e^{-at} u(t) \rightleftharpoons \frac{1}{s+a} \; ; R$$

$$t e^{-at} u(t) \rightleftharpoons \frac{1}{(s+a)^2} \; ; R$$

$$\frac{t^2}{2} e^{-at} u(t) \rightleftharpoons \frac{1}{(s+a)^3} \; ; R$$

$$\vdots \qquad\qquad \vdots$$

$$\frac{t^{n-1} e^{-at} u(t)}{(n-1)!} \rightleftharpoons \frac{1}{(s+a)^n} \; ; R$$

Properties

① Convolution in time domain
≡ Multiplication in s-domain

$x(t) \rightleftharpoons X(s)$ ROC: R_1

$h(t) \rightleftharpoons H(s)$ ROC: R_2

$y(t) = x(t) * h(t) \rightleftharpoons X(s)H(s) = Y(s)$
ROC: Including $R_1 \cap R_2$

let $X(s) = \frac{s+1}{s+2}$

$H(s) = \frac{s+2}{s+1}$

$Y(s) = X(s)H(s) = 1$

$\boxed{y(t) = \delta(t)}$ s-plane

can be larger

ROC ≡ entire

② Shifting in the s-domain

$x(t) \rightleftharpoons X(s)$ ROC: R

ROC: $\mathrm{Re}\{s\} > -a$

$e^{s_0 t} x(t) \rightleftharpoons X(s-s_0)$

$\Downarrow$

$\mathrm{Re}\{s-s_0\} > -a$

$\mathrm{Re}\{s\} > -a + \mathrm{Re}\{s_0\}$

ROC : $R + \mathrm{Re}\{s_0\}$

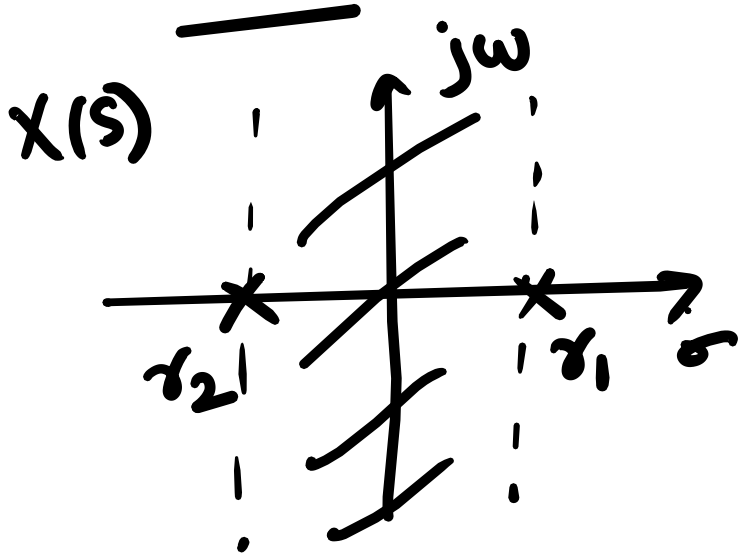
③ Time Scaling

$x(t) \rightleftharpoons X(s)$ ROC : R

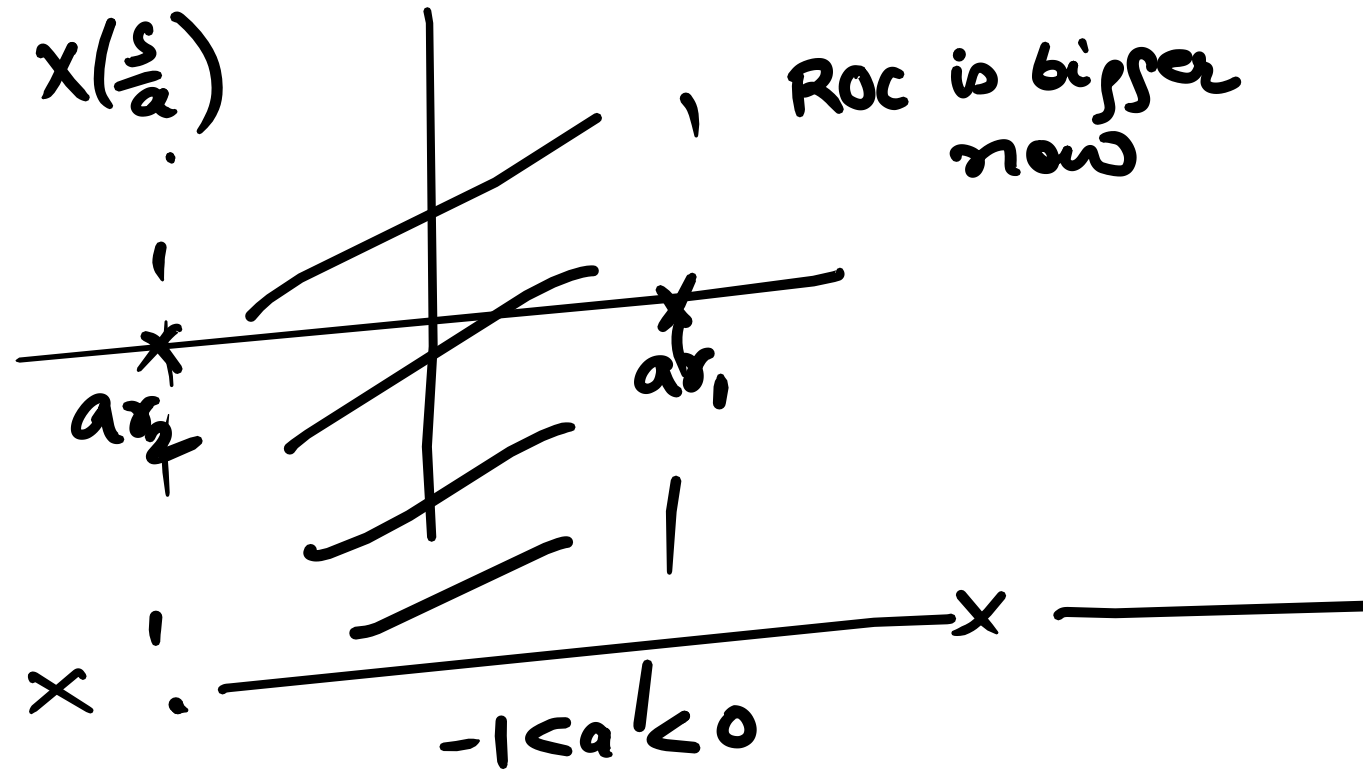
$x(at) \rightleftharpoons \frac{1}{|a|} X\left(\frac{s}{a}\right)$ ROC : aR

$a > 0$

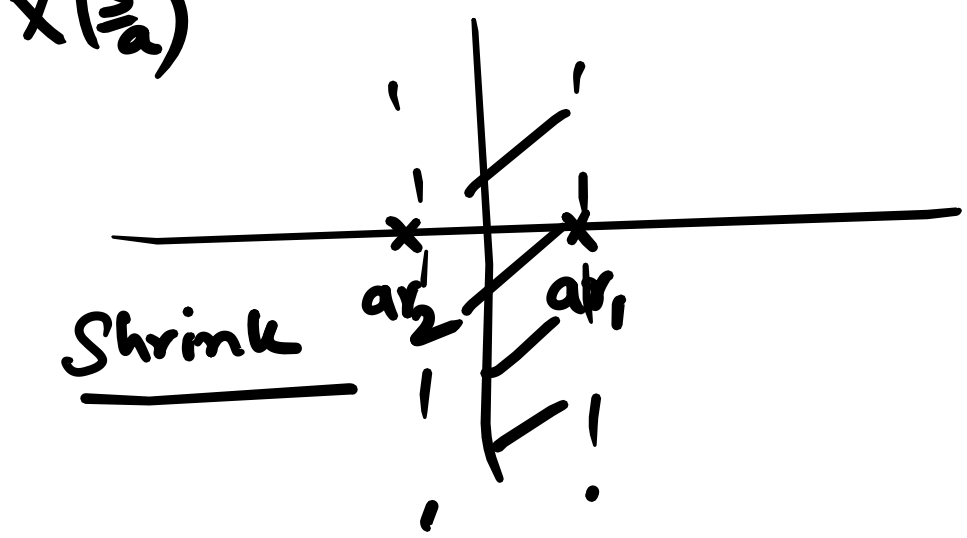
$a > 1$



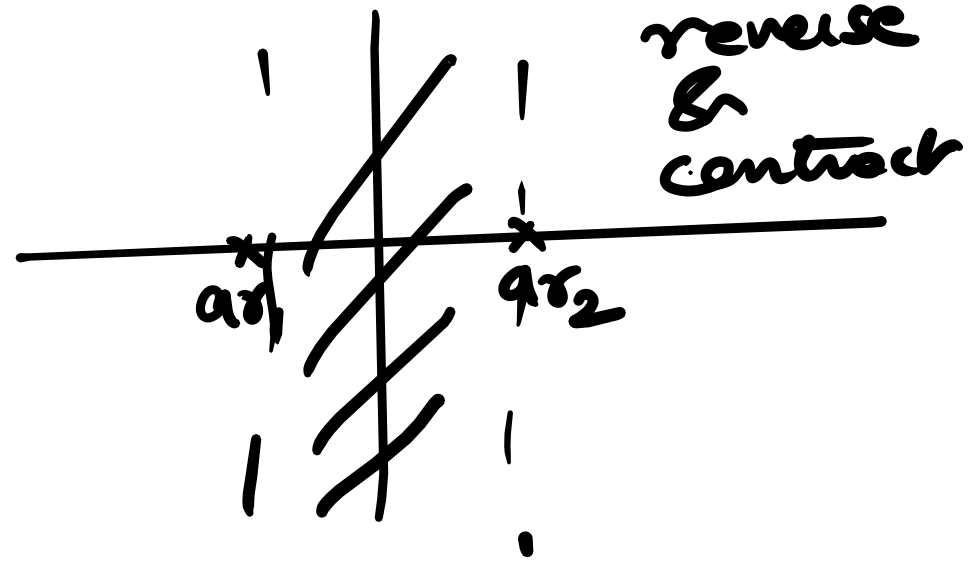
$0 < a < 1$



$-1 < a < 0$



Shrink



④ Differentiation in time-domain

$$x(t) \rightleftharpoons X(s) \quad \text{ROC: } R$$

$$\frac{d}{dt}x(t) \rightleftharpoons sX(s) \quad \text{ROC contains } R.$$

⑤ Integration in time-domain

$$x(t) \rightleftharpoons X(s) \quad \text{ROC: } R$$

$$\int_{-\infty}^{t} x(\tau)\,d\tau \rightleftharpoons$$

$$= x(t) * u(t) \rightleftharpoons X(s)U(s) = \frac{X(s)}{s}$$

at least $R \cap \{Re\{s\} > 0\}$

⑥ Initial & Final value theorems —

Ⓐ if $x(t) = 0$ for $t < 0$

Ⓑ & $x(t)$ contains no impulses or higher order singularities at $t = 0$,

then $$x(0^+) = \lim_{s \to \infty} s\,X(s)$$ Initial value theorem

& $$\lim_{t \to \infty} x(t) = \lim_{s \to 0} s\,x(s)$$ Final value theorem

Ex $x(t) = 3e^{-2t}u(t) - 2e^{-t}u(t)$

$$X(s) = \frac{3}{s+2} - \frac{2}{s+1} \quad \text{Re}\{s\} > -1$$

$$X(s) = \frac{s-1}{(s+1)(s+2)}$$

$$x(t)\Big|_{t=0^+} = 1$$

$$\lim_{s\to\infty} sX(s) = \lim_{s\to\infty} \frac{s(s-1)}{(s+1)(s+2)}$$

$$= \lim_{s\to\infty} \frac{1 - 1/s}{1 + 3/s + 2/s^2} = \frac{1}{1} = 1$$

$\lim_{t \to 0} x(t) = 0$

$\lim_{s \to 0} sX(s) = \lim_{s \to 0} \frac{s(s-1)}{(s+1)(s+2)}$

$= \frac{0}{2} = 0$

Verified the Final value theorem

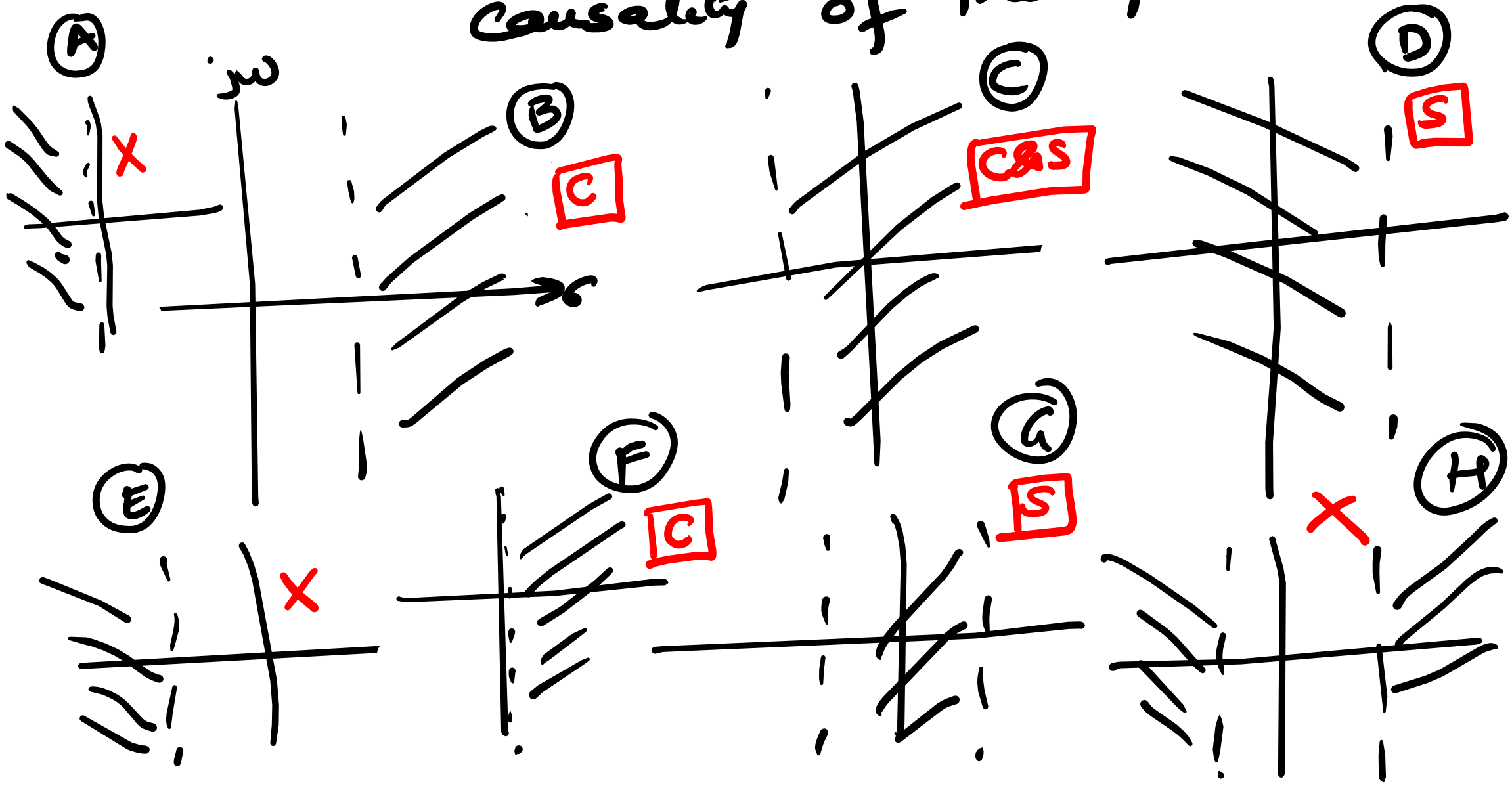
LTI Systems & analysis using L.T.

① System is stable $\longrightarrow$

$$\int_{-\infty}^{\infty} |h(t)|\,dt < \infty$$

Given H(s) & ROC of H(s) Is H(s) is rational

Can we comment on the stability & causality of the system.



An LTI system is characterized by linear constant-coefficient differential eq.

$$\frac{d}{dt}y(t) + \alpha y(t) = \beta x(t)$$

$$H(s) = \frac{Y(s)}{X(s)}$$

$$sY(s) + \alpha Y(s) = \beta X(s)$$

$$H(s) = \frac{\beta}{s+\alpha}$$

$$\mathrm{Re}\{s\} > -\alpha$$

or $\mathrm{Re}\{s\} < -\alpha$

if $\quad Re\{s\} > -\alpha$

$$h(t) = \beta e^{-\alpha t} u(t)$$ Causal System

if $\quad Re\{s\} < -\alpha$

$$h(t) = -\beta e^{-\alpha t} u(-t)$$

x

Ex Consider an LTI system with the following properties

(i) The system is causal

(ii) The system fn. is rational & has only two poles at $s=-2$ & $s=-4$

(iii) if $x(t)=1$, then $y(t)=0$

(iv) The value of the impulse fn. at $t=0^+$ is 4

Find $H(s)$.

(i) $h(t)$ is right sided. ROC of $H(s)$ is right to the rightmost pole.

(ii) $H(s) = \dfrac{N(s)}{(s+2)(s+4)}$

$H(0)=0$

(iii) if $x(t)=e^{st}$, $y(t) = H(s)e^{st}$

if $s=0$, $x(t)=1 \rightarrow y(t)=0 \Rightarrow H(s)\big|_{s=0}=0$

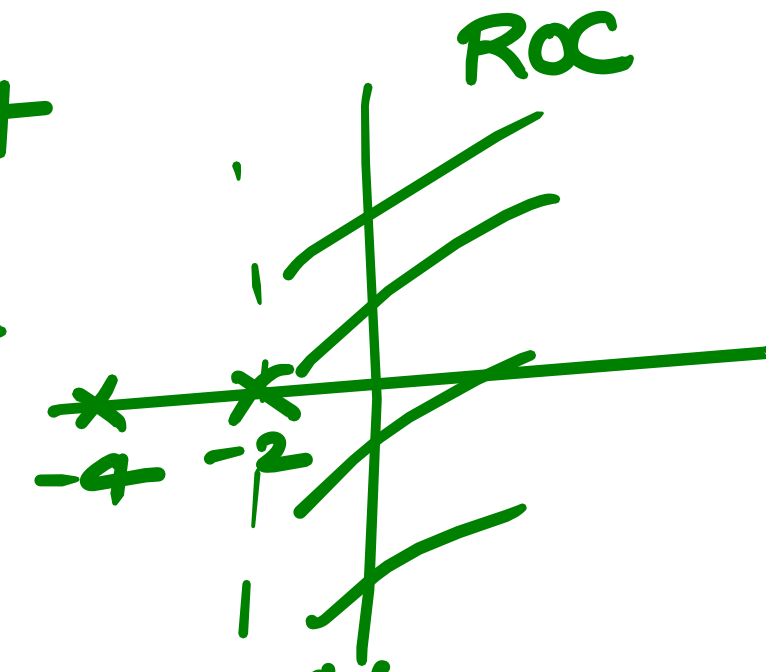
$\Rightarrow$ zero is a zero at $s=0$

$$\therefore \quad H(s) = \frac{N(s)}{(s+2)(s+4)} = \frac{sq(s)}{(s+2)(s+4)}$$

(iv) $$R(t)\Big|_{0+} = 4 \Rightarrow \lim_{s\to\infty} sH(s) = 4$$

$$\lim_{s\to\infty} \frac{s^2 q(s)}{(s+2)(s+4)} = 4$$

$$\Rightarrow \quad q(s) = 4$$



$$\therefore \quad H(s) = \frac{4s}{(s+2)(s+4)}$$

$$h(t) = -4e^{-2t}u(t) + 8e^{-4t}u(t)$$

Ex $X(s) = \dfrac{2s^2+5s+5}{(s+1)^2(s+2)}$ $\mathrm{Re}\{s\} > -1$

Find $x(t)$.

$$x(t) = 2te^{-t}u(t) - e^{-t}u(t) + 3e^{-2t}u(t)$$

Ex $h(t) = e^{-t}u(t)$ ×

$H(s) = \dfrac{1}{s+1}$ ROC : $\mathrm{Re}\{s\} > -1$

Causal system.

Ex let $H(s) = \frac{e^s}{s+1}$ $\quad Re\{s\} > -1$

$$h(t) = e^{-(t+1)} u(t+1)$$

$h(t) \neq 0$ for $t < 0$

System is not causal.

Conclusion if a system is causal, ROC is right to the rightmost pole.

Converse is true only if $H(s)$ is rational.